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Hedonic Price Analysis in Food Markets

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Abstract and Keywords

This article presents the basic theory of hedonic modeling, its empirical application and relevance, and the principal limitations and challenges. Agricultural economists have long utilized the hedonic price relationship, and the hedonic price technique has been utilized to estimate the implicit prices of attributes for numerous food products. It provides a framework of the hedonic models that can be used to quantify the long-term effects of exogenous shocks on quality, or implementation of regulatory policies. This article discusses the whole armamentarium of econometric models and methods familiar to applied economists, including parametric, semi-parametric, and non-parametric approaches. The objective is to describe briefly each econometric model within the hedonic context and provide an understanding of the implications and trade-offs inherent to the choice of each alternative. This is a fruitful area of research with increased customization of products and the increasing availability of large data sets.

Keywords: hedonic modeling, hedonic price relationship, econometric models, trade-offs, regulatory policies

1 Introduction

A transformation has been occurring in modern food markets. Consumer expectations for quality are increasing and, at the same time, consumers demand increasingly customized products. One can think about the common habit of ordering an espresso drink. Coffee used to come with two options: cream and sugar. Now there are thousands of combinations, and consumers are willing to pay for them. Many simple commodity markets have evolved into highly differentiated product markets in order to fulfill heterogeneous consumer preferences for product attributes (Barkema and Drabenstott 1995). Customization contributes to consumer welfare because it enables each consumer to select an attribute bundle that comes as close as possible to matching her preferences. In addition to a larger number of product attributes, greater variety within each one of them increases the likelihood that she might obtain exactly the option that maximizes her utility. This customization of purchases has implications for economic theory and estimation, especially how researchers analyze markets. Approaches that perform well for examining food commodities have often limited applicability and usefulness when studying differentiated food products.

Traditional demand theory is based on the premise that a product generates utility. Hence, utility theory has been used to analyze consumers' choice and consumption of a good based on price and a budget constraint. In economic models of demand for differentiated products, consumer utility is derived from the summation of utilities associated with each individual attribute embedded in the product (Lancaster 1966; McFadden 1973; Rosen 1974; Berry, Levinsohn, and Pakes 1995).

Even though the origin of the study of differentiated products, and specifically hedonic methods, is often traced to the work of Waugh 1928 and the field of agricultural and food economics, the discipline has impacted numerous fields of economics. Important developments in the theory and empirics of hedonics have come from the consumer price index literature (see Triplett 2006 for a summary), real estate economics, non-market valuation of environmental amenities (see Freeman 1993; Taylor 2003), and hedonic modeling of labor markets (see Viscusi 1993). Many of the findings in these areas of application have obvious implications for performing hedonic studies on food products. In this chapter we present the basic theory of hedonic modeling, its empirical application and relevance, and the principal limitations and challenges.

2 Hedonic Theory

Hedonic price analysis is based on the hypothesis that consumers' utility is not generated by the purchased product itself, but rather by the qualities and characteristics it contains. Differentiated goods are therefore treated as bundles of various quality attributes which differentiate them from other related goods, and the observed equilibrium market price is a function of the (implicit) prices of each quality attribute. This idea is formally presented in Lancaster 1966, but the earliest hedonic-type models can be traced back to the contributions of early agricultural economists. Based on the observation that vegetable lots in Boston produce markets showed

considerable variations in price, Waugh 1928 regressed the price of asparagus on color, size of stalks, and uniformity of spears to determine which quality traits significantly influenced prices. Similarly, Court 1939 analyzed the characteristics influencing automobile prices. In the 1960s the relationship between product characteristics and prices was examined in order to separate the effect of changes in product quality from inflation (Adelman and Griliches 1961; Griliches 1971).

Early contributions to the formal theory of hedonic pricing theory include Court 1941, Tinbergen (1951), and Rosen 1974, who is often referenced as the seminal contribution. Even though the principal original contribution of the latter article relates to hedonic demand estimation, Rosen 1974 formalized how the hedonic price function, mapping multiple product attributes into price space, can be obtained from the market behavior of profit-maximizing firms and utility-maximizing consumers. Our presentation of the theoretical framework closely follows his work. Other useful presentations of hedonic price theory in an environmental economics context include Freeman 1993 and Taylor (2003).

2.1 The Hedonic Model

Consider a row vector of K product characteristics identifying the i th product, $z_i = (z_1, \dots, z_K)$, and assume that there are a significantly large number of differentiated products to ensure that a "spectrum of products" exists, allowing for marginal analysis. In the remainder of this chapter, we use the sub k to indicate or emphasize an attribute, and i to indicate a product. Vectors appear in bold, and subs are omitted when the context is not ambiguous. Each differentiated product is offered at a posted market price depending on its vector of attributes, so that market prices can be represented by the (unknown to the researcher) *hedonic price function* $p(z_1, \dots, z_K)$. The key assumptions of Rosen's development of the hedonic model are largely collinear to those of perfect competition: all consumers are aware of all the available versions of a differentiated product, switching from one product to another is costless, and a large number of producers and consumers with no market power buy and sell the differentiated product. Even though we adopt these assumptions in the following exposition, we emphasize that the existence of a hedonic price function taking the form $p(z)$ arises from consumers' preferences for product attributes, and does not hinge on the assumption of perfect competition (Bajari and Benkard 2005). Indeed, modified versions to account for firms' market power or non-continuous product space have been developed (see Pakes 2003; Bajari and Benkard 2005).

The utility a consumer obtains from consuming good $z = (z_1, \dots, z_K)$ is represented by $U = U(x; z_1, \dots, z_K; \alpha)$, where x is a composite good normalized to have a unit price, and α is a taste parameter. Maximizing utility subject to the budget constraint $y = x + p(z)$, where y is income, results in the first-order condition $\frac{\partial p}{\partial z_k} = p_{z_k} = \frac{U_{z_k}}{U_x}$, $k = 1, \dots, K$. This set of conditions implies that utility is maximized when a consumer is indifferent between switching to a product with an extra unit of any attribute and paying the associated implicit price p_{z_k} , or using the money to purchase the composite good. The second-order conditions will be satisfied with the usual assumptions on utility and if $p(z)$ is not too concave. The *bid function* $\theta = \theta(z_1, \dots, z_K; u, y, \alpha)$ captures the amount that a consumer is willing to pay for the vector of characteristics z as a function of a fixed level of utility u , income, and consumer's taste. This function is made to satisfy the relationship (1)

$$U(y - \theta, z_1, \dots, z_K; \alpha) = u,$$

thereby defining a family of indifference surfaces relating the levels of product characteristics and their valuation in terms of forgone money (in the form of the amount of consumption of the composite good $y - \theta$). Differentiation shows that the bid functions are increasing in z_k , i.e., $\theta_{z_k} = \frac{\partial \theta}{\partial z_k} > 0$ for all consumers, yet they may differ across consumer with different income or tastes.

Consider now the producers' decisions, which is symmetric to the consumer's problem. Following again Rosen's notation, let $M(z)$ represent the number of goods produced with product characteristics z , and $C(M, z; \beta)$ indicate the cost function, where the parameters β differentiate firms based on the factor prices they face and their production technology. Assume that the cost function is increasing and convex in M and z . Each firm maximizes profits $\pi = Mp(z) - C(M, z; \beta)$ by optimizing with respect to the choices of M and z . At the optimum, the marginal revenue from an additional unit of z is equal to its marginal cost of production, $p(z) = C_M$, and the marginal revenue from increasing any product attribute is equal to the (per unit) marginal cost of adding it to the bundle:

$p_{z_k}(z) = \frac{C_{z_k}}{M}$. Analogous to the consumer's bid function, the seller's *offer function* $\phi = \phi(z_1, \dots, z_K; \pi, \beta)$ indicates the price that a firm characterized by the vector β is willing to accept for selling the differentiated good z and maintain the (fixed) profit π . This function defines a family of producer indifference surfaces in price characteristics space (given that M is optimally chosen), and satisfies the necessary relationships (2)

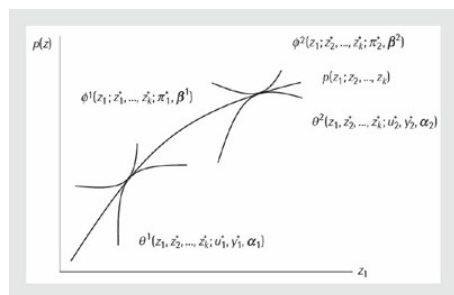
$$\pi = M\phi - C(M, z_1, \dots, z_K; \beta) \text{ and } C_M(M, z_1, \dots, z_K; \beta) = \phi.$$

Given convexity of the cost function in characteristic space, the per unit marginal implicit costs of increasing each characteristic, $\phi_{z_k} = \frac{C_{z_k}}{M}$, $k = 1, \dots, K$, is positive. Therefore, all firms' offer functions are increasing in the levels of z , but may differ depending on technology and factor prices.

Since there are many consumers and supply is atomistic, consumers and producers take the price function $p(z)$ as given. Thus, using asterisks to indicate optima, a consumer's utility is maximized when $\delta(z^*; u^*; y; \alpha) = p(z^*)$ and $\theta_{z_k} = p_{z_k}(z^*)$, $k = 1, \dots, K$, i.e., when the bid functions are tangent to the hedonic price function in all the dimensions of z . Analogously, maximum profit and an optimal mix of attributes imply that $p(z^*) = \theta(z_1^*, \dots, z_K^*; \pi^*, \beta)$ and $p_{z_k}(z^*) = \phi_{z_k}(z_1^*, \dots, z_K^*; \pi^*, \beta)$, $k = 1, \dots, K$.

When the conditions $\theta_{z_k}(z^*) = \phi_{z_k}(z^*) = \theta_{z_k}(z^*)$, $k = 1, \dots, K$; and $p(z^*) = \theta(z_1^*; u^*; y; \alpha) = \theta(z_1^*, \dots, z_K^*; \pi^*; \beta)$ hold for at least one

consumer and a producer, a sale occurs. In other words, observing a market transaction for the product z_i at price $p(z_i)$, the pair $\{p(z_i); z_i\}$ implies that the marginal bid of the buyer and the marginal offer of the seller are exactly matched, and equal the implicit marginal hedonic price function determined by the market in all the dimensions of the vector z . This is depicted in Figure 6.1, which shows tangencies between two producer-consumer pairs (assuming strictly concave bid and convex offer functions), and is similar to Rosen's (1974) and Taylor's (2003) representations.



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Figure 6.1 Hedonic price equilibrium

The figure also portrays how the hedonic price function is found by tracing the locus of tangency points between the buyer's bid and the seller's offer curve, and $p(z_i)$ represents a joint envelope of a family of value functions. Given a stable distribution of preferences and technology in the population of consumers and firms participating in a market (i.e., in cross-sectional data), the hedonic price function can be represented as being solely a function of product attributes: (3)

$$p(z) = p(z_1, z_2, z_3, \dots, z_K)$$

Thus, once product attributes and prices are observed, forming the data set $\{p(z_i); z_i\}$, $i = 1, \dots, N$, a researcher can empirically determine the shape of the hedonic price function $\hat{p}(z)$ and recover marginal implicit prices for each attribute $\hat{p}_{zk}(z)$, $k = 1, \dots, K$. This task is often referred to as the "first stage" of hedonic regression. Figure 6.1 portrays an application in which the hedonic price function is concave in z_1 and thus the implicit price of the attribute z_1 (the market price of adding an extra unit of z_1 to the bundle) diminishes as z_1 grows larger. This need not always be the case and the hedonic price function can take virtually any shape. Indeed, the principal empirical challenge (which we discuss in Section 3) of the first-stage regression stems from the lack of guidance from hedonic theory in choosing a functional form. What can perhaps be inferred from the setup of the model is that, given a population of heterogeneous consumers and firms, the hedonic price functions and its derivatives are likely non-linear in z , and therefore implicit marginal prices $\frac{\partial \hat{p}(z)}{\partial z_k} = \hat{p}_{zk}(z)$ are a function of the vector of attributes embedded in a product.

As mentioned before, if one observes the bundle of characteristics that is actually chosen by a given consumer, then the set of implicit prices $\hat{p}_{zk}(z^*)$, $k = 1, \dots, K$ (evaluated at the chosen level of attributes), represents his/her willingness to pay for each z_k , as implied by the consumer optimality conditions. Taylor (2003: 263) shows how, based on this general idea, the estimated hedonic price function can be used to perform some limited welfare computations. We emphasize, nevertheless, that marginal implicit price functions should not be directly interpreted as general measures of consumers' willingness to pay for product attributes. Since both cost and willingness to pay factors determine the implicit prices, their interpretation is closer to that of an equilibrium market price. To provide an example, it is incorrect to infer that, because a particular attribute has the largest implicit price, it is the attribute that consumers value the most. High implicit prices may very well be due to elevated costs of production, and it is possible that only a small fraction of consumers actually purchase bundles containing that expensive attribute.

The *second stage* of hedonic regression refers to the estimation of the aggregated demand for (and supply of) individual product attributes, which allows for classical welfare analysis and comparison of alternative policies. Food-related examples include laws regulating the availability of product attributes or mandating their minimum or maximum levels (e.g., trans-fat oils, fat content, etc.). Unfortunately, the estimation of hedonic demand and supply functions is complicated by several empirical challenges, notably identification of demand and supply parameters and endogeneity of the second-stage regressors. Even though the debate over the empirical issues of hedonic demand estimation has evolved in a joint discussion, here we try to separate the two principal issues for the sake of exposition.

As in all demand studies, simply observing all the pairs of (implicit) prices and quantity purchased $\{p_{zk}(z^*), z^*\}$, $k = 1, \dots, K$, does not allow identifying the demand or supply schedules for each attribute. To estimate demand, we need variables that are known to shift demand (but not supply), and are not included in the first-stage regression. Along these lines, Rosen suggested that, once estimated implicit marginal price functions are used to calculate attributes' prices at the level purchased by each consumer, aggregate inverse demand (D) and supply (S) for the k th attribute could be recovered by regressing $\hat{p}_{zk}(z^*)$ against y_1 , a vector of product attributes and a vector of sociodemographic consumer characteristics (the empirical counterpart of a), or a vector of producers' characteristics, y_2 , yielding (asterisks omitted from now on): (4)

$$\hat{p}_{zk}(z) = D(z, y_1)$$

(5)

$$\hat{p}_{zk}(z) = S(z, y_2)$$

One pitfall of this approach, which was pointed out by Brown and Rosen 1982, is that if the $\hat{p}_{zi}(z)$ are linear combinations of z (say, $Z(Z'Z)^{-1}Z'p$ from a first-stage linear regression in which Z is $N \times K$ and p is $N \times 1$), equations (4) and (5) are still not identified. As Brown and Rosen wrote, "Any new information that they [the estimated implicit prices] may provide can only come from a priori restrictions placed on the functional form of the hedonic price function" (1982: 766).

Indeed, to overcome the difficulty, some authors (e.g., Quigley 1982) imposed additional identifying structure on the hedonic model via specific assumptions on the functional form of consumers' utility. Palmquist 1984 and Epple 1987 took another avenue by proposing the use of data from multiple markets as a source of identifying information. The logic is that, if the equilibrium hedonic price functions differ across markets but tastes can be controlled for, hedonic functions estimated for multiple markets allow obtaining several $\{p_{zk}(z), z\}$ pairs for each "type" of consumer.

After a long hiatus in the debate over identification, Ekeland, Heckman, and Nesheim (2002, 2004) and Bajari and Benkard 2005 revived the discussion. Bajari and Benkard 2005 identified demand by assuming that the parametric form of utility is known, but leave the distribution of utility parameters across consumers unrestricted. Ekeland, Heckman, and Nesheim (2002) proposed a promising approach that exploits the recent advances in non-parametric methods and availability of computing power. Their method is based on extracting, rather than assuming, the identifying information embedded in the shape of the hedonic price function via non-parametric estimation.

The second major problem of hedonic demand and supply estimation relates to the endogeneity of the second-stage regressors. Brown and Rosen 1982 and Bartik 1987 noted that, since marginal implicit prices usually depend on the level of product attributes, consumers' choices of the pairs $\{p_{zk}(z), z\}$ are simultaneous and perhaps correlated with unobserved taste variables (and thus the error term), biasing second-stage estimates. Some authors refer to this issue as the "sorting" problem. Consistent estimation of the demand and supply parameters can in theory be achieved via traditional two-stage least squares techniques, yet appropriate instruments for the endogenous regressors need to be available. In the words of Bartik, "The practical problem for empirical hedonic research [i.e., estimating hedonic demand parameters] is finding instruments whose exogeneity can be defended with some plausibility" (1987: 87).

Owing to the novelty and complexity of the recent non-parametric approaches to identification and the difficulties of finding good instruments or multi-market data sets, empirical estimation of second-stage hedonic demand models has been limited, especially in food markets. An exception is Nerlove 1995, who noted that in certain circumstances prices can be considered exogenous, and thus demand is identified and estimable in a single stage if data on quantity purchased of each differentiated good is available. He considered the wine market in Sweden, the argument being that Sweden represents a very small fraction of the world market, has no domestic production, and wine prices are controlled by the government, which simply adds a percentage markup to import prices. To offer here a rather simplified version of his model, Nerlove considers a whole class of closely related (but differentiated) commodities, and the representative consumer chooses a spectrum of quantities $Q(z_i)$, $i = 1, \dots, N$, to consume¹ based on their income and preferences. The empirical model arising from utility maximization takes the form (6)

$$Q(z) = R[p(z), z, y_1],$$

The empirical usefulness is that, *when supply is completely elastic and prices are exogenous*, one can estimate demand for each bundle or variety by regressing the quantities of each product against its unit price, the measures of quality attributes characterizing the variety, and consumers' characteristics.

Obviously, under this specification the coefficients associated with the quality characteristics z do not capture the implicit prices calculated in Rosen's traditional setting. Rather, they measure the difference in quantity consumed that can be linked to the presence of an extra unit of a specific product attribute, holding price constant. Implicit prices, the market price for a marginal increase in one attribute holding quantity constant, can nevertheless be recovered from (6) via the implicit function theorem as $p_{zk} = \frac{R_{zk}}{R_p}$. In summary, the method applies to settings in which consumers' choices do not affect prices and supply is virtually unlimited (see Durham, Pardoe, and Vega 2004 for an application to wine choices in restaurants).

Even when estimation of demand and supply of attributes is not possible because of the rather cogent technical and data requirements, simple estimation of the hedonic price function provides a wealth of information. Indeed, a wide array of empirical results are owed to first-stage regression (e.g., mass appraisal of real estate prices based on house characteristics, the computation of price indices separating the price effect of quality improvements from inflation, the estimation of the market value of environmental amenities or the reputation of firms).

2.2 Hedonic Modeling of Food Products

As discussed in the previous section, agricultural economists have long utilized the hedonic price relationship, and the hedonic price technique has been utilized to estimate the implicit prices of attributes for numerous food products, creating a rather vast literature. Again, most studies are limited to the (first-stage) estimation of the hedonic price functions. Examples include estimating the implicit price of apple size and grades (Tronstad, Huthoefer, and Monke 1992; Carew 2000); tuna qualities (McConnell and Strand 2000); nutritional and convenience factors in breakfast cereals (Stanley and Tschirhart 1991); chemical composition of wheat (Espinosa and Goodwin 1991); the value of leaner cuts (Unnevehr and Bard 1993) and package size and brand name in beef (Ward, Lusk, and Dutton 2008) and pork (Parcell and Schroeder 2007); and the price premium for organic tomatoes (Huang and Lin 2007) or veal

labeled with specific geographical indication of origin (Loureiro and McCluskey 2000).

A large literature of hedonic models has been developed for wine, largely because of the considerable product differentiation in the industry. Combris, Lecocq, and Visser (1997) showed that objective characteristics (such as expert rating score and vintage) are statistically significant in the hedonic price equation, while sensory variables (such as tannins content and other measurable chemicals) are not. This is not to say that quality has nothing to do with wine prices. Many researchers (Oczkowski 1994; Landon and Smith 1997; Schamel and Anderson 2003) have found that expert quality ratings are significant explanatory variables. Possible explanations for the insignificance of sensory cues are the consumers' difficulty with isolating the effect of each chemical on the final flavor and smell and that only a small percentage of wine purchasers are connoisseurs. In summary, expert ratings act as a quality signal to the consumer.

Since consumers cannot assess wine quality until after consumption, but they are aware that a large range in quality and prices exists, the applications of the hedonic method to the wine market offers an excellent framework to study reputation effects. Numerous studies (e.g., Schamel and Anderson 2003; Steiner 2004; Costanigro, McCluskey, and Mittelhammer 2007; Carew and Florkowski 2010) have found that certain regions of production command a *ceteris paribus* premium, which captures production costs differentials, comparative advantages due to terroir, and the value of the collective reputation of a production district. Landon and Smith 1998 used a hedonic model to explicitly model reputations as consumers' expectations on quality, and separated firm (winery) and regional reputations. Extending this line of research, Costanigro, McCluskey, and Goemans (2010) used a hedonic wine model to study how names and reputations nest within each other, and showed that reputation price premiums tend to migrate from aggregate (regional) names to specific (firm) ones as product prices increase.

To summarize, first-stage estimates of implicit prices of food attributes are useful on several accounts. First, producers aware of their own production costs can use implicit prices to devise marketing strategies and optimal mix of attributes. Furthermore, the hedonic approach is an effective tool in isolating the premium for "credence attributes" such as organic certification, geographical indications, or other socially responsible characteristics (e.g., dolphin-friendly tuna, fair trade coffee) that consumers cannot verify even after purchase. The added policy and regulatory interest of such studies originates from the producers' incentives when credence attributes have high implicit prices. In the absence of regulations, some producers may make a false claim and pocket the premium at zero production cost. Similarly, production districts with positive price premiums will want to prevent other producers from unduly using their name (e.g., the "Champagne" wine case), or their own members from lowering quality to save on production costs and free-ride on collective reputation (see Winfree and McCluskey 2005). Within this framework, hedonic models can be used to quantify the long-term effects of exogenous shocks on quality, or implementation of regulatory policies.

3 Hedonic Empirical Issues

The relationship between prices and attributes is left unspecified by hedonic theory. Even the additive separability of the product attributes in the hedonic price function, a common tenet of most empirical hedonic applications, is an econometric assumption rather than a result derived from theory. It is generally accepted that the appropriate econometric model and functional form should be chosen by the researcher based on the specific application and the available data.

Owing to this lack of theoretical structure, a survey of the work accomplished since the early days of hedonic modeling encompasses the whole armamentarium of econometric models and methods familiar to applied economists, including parametric, semiparametric, and non-parametric approaches. Our objective is to describe briefly each econometric model within the hedonic context and provide the reader with an understanding of the implications and trade-offs inherent to the choice of each alternative. The focus is kept on the first stage of hedonic regression.

3.1 Parametric and Semi-Parametric Models

In its most basic specification the estimated hedonic function takes the following linear form: (7)

$$P_i = \alpha_0 + \sum_{k=1}^K \alpha_k z_{ki} + \varepsilon_i;$$

while additivity and linearity in parameters are common assumptions of parametric models, the chosen functional forms vary widely and often include non-linear transformations of the dependent variable, the regressors, or interaction. Ekeland, Heckman, and Nesheim (2002, 2004) emphasize that, since the curvature of the hedonic function is determined by the curvature of heterogeneous firm profit functions and consumers' utility functions, hedonic and implicit price functions can be expected to be non-linear. Regarding the parametric distribution of product prices little can be said a priori, but many authors note that non-negativity and right-skewness are features consistent with the log-normal distribution.² A logarithmic transformation of the dependent variable in (7) therefore yields (8)

$$\ln(P_i) = \alpha_0 + \sum_{k=1}^K \alpha_k z_{ki} + \varepsilon_i$$

where ε is distributed $N(0, \sigma^2)$ if prices are indeed log-normally distributed. A survey of the literature will reveal that the log-linear model is by far the most commonly used specification in applied hedonic models and it is considered to be the "canonical" functional form in certain applications (Aldy and Viscusi 2008). Despite this popularity, it is not uncommon for empirical tests to reject the log-linear model

in favor of alternative specifications (e.g., Halvorsen and Pollakowski 1981; Landon and Smith 1998), and generally the adoption of a specific functional form on a priori grounds is not justified.

Specification searches can be implemented by estimating several alternative functional forms. After their parallel estimation, alternative econometric models are measured against a series of specification (e.g., the Ramsey Regression Equation Specification Error Test (RESET) test by Ramsey 1969) and other diagnostic tests assessing homoskedasticity (Breusch-Pagan; Goldfeld-Quandt) or normality (Shapiro-Wilk, Jarque-Bera), to choose the “best” model (see Landon and Smith 1998 for an example). Out-of sample forecasting performance should also be an important choice criterion for selecting an appropriate specification, as evaluation of within-sample performance alone can produce overfitted models and non-robust results. This can be accomplished by setting aside a small subsample (say, 10 percent of the available observations) for model validation purposes (for example, Fletcher, Mangan, and Raeburn 2004 consider mean forecasting error and mean percent forecasting error).

The single-parameter Box-Cox transformation of the dependent variable (Box and Cox 1964) has seen many applications in the hedonic literature (e.g., Stanley and Tschirhart 1991; McCluskey and Rausser 2003). The model takes the form (9)

$$p_i^\theta = \alpha_0 + \sum_{k=1}^K \alpha_k z_{ki} + \varepsilon_i$$

where $p_i^\theta = \frac{(p_i^\theta - 1)}{\theta}$ for $\theta \neq 0$, and $p_i^\theta = \ln(p_i)$ for $\theta = 0$. Estimation involves simultaneously finding the transformation of the dependent variable, θ , and other model parameters maximizing a likelihood function³ derived from a normality assumption. Thus, Box-Cox procedures simultaneously pursue normalization of the dependent variable (and the error term) and best possible fit. While the procedure was originally designed as a way of imposing approximate normality on a non-normally distributed random variable, in the hedonic context it is often used to determine the appropriate choice of functional form. For example, a failure to reject the null hypothesis that $\theta = 0$ results in the adoption of a log-linear model. Even though this practice is widespread and generally accepted, we point out that normalization was a historical necessity when the only available tool was least squares estimation (which is sensitive to the outliers typical of skewed distributions). Nowadays, a variety of estimation methods robust to outliers are available, and the need for normalization seems less cogent.

Halvorsen and Pollakowski 1981 extended the use of this approach to create the quadratic Box-Cox functional form, a nesting approach in which an extremely flexible functional form allows deriving many specifications as special cases. The model takes the form (10)

$$p_i^\theta = \alpha_0 + \sum_{k=1}^K \alpha_k z_{ki}^\lambda + \frac{1}{2} \sum_{k=1}^K \sum_{j=1}^K \gamma_{kj} z_{ki}^\lambda z_{ji}^\lambda + \varepsilon_i$$

where $\gamma_{kj} = \gamma_{jk}$ and θ and λ are Box-Cox transformations. Many alternative functional forms, including the translog (Christensen, Jorgenson, and Lau 1973), log-linear, double log, and quadratic, can be compared by testing the restrictions imposed on (7) by each functional form. Modern software packages often allow estimating a value of λ_k specific to each attribute. When this is done, interaction terms can usually be omitted.

While current econometric software makes it relatively simple to estimate such models, their cost may be prohibitive in terms of degrees of freedom, convergence issues, induced collinearity, and, most importantly, the overall interpretability of the model (this point was stressed by Cassel and Mendelsohn 1985). In addition to the obvious relationship between dependent and independent variables and the resulting implicit price function, the implications of choosing one functional form over another can also be analyzed by considering the implied hedonic contours. Hedonic contour graphs portray the possible combinations of any two attributes yielding the same product price, holding the other attributes constant (see Triplett 2006: 175).

3.2 Heteroskedasticity

Heteroskedastic errors are widespread in hedonic applications, and specific patterns of heteroskedasticity may be typical of a specific good or group of products. Often, the variance of the error term increases with the magnitude of the predicted prices,⁴ yielding the heteroskedastic process $\sigma_i^2 = \sigma^2(\hat{p}_{ik})$, where σ^2 is a constant (see Faux and Perry 1999 in a valuation of irrigation water for an example). One possible explanation for this increase in the variance of prices about the conditional mean is provided by Rosen's (1981) article “The Economics of Superstars.” In other instances, the variance may be related to a product attribute (or more), taking the form $\sigma_i^2 = \sigma(z_{ik})$. To provide an example, building-age-related heteroskedasticity is common in real estate hedonic models (Goodman and Thibodeau 1998).

Under heteroskedasticity, the OLS estimator is inefficient and the standard estimator of the variance-covariance matrix is biased. The hedonic literature has dealt with the problem adopting the classical textbook remedies: one can either rely on the consistency of the OLS estimator while correcting the estimates of the coefficients' standard errors (White 1980) or adopt a weighted least squares approach to transform the data. It is worth reminding that the second approach implies knowing or discovering the correct structure of heteroskedasticity at the risk of biasing the coefficient estimates, and researchers should keep in mind the Hippocratic precept “First, do no harm” when dealing with this problem.

Since heteroskedasticity can be induced by misspecification, certain data transformations may stabilize the variance. It is important to

discriminate between induced heteroskedasticity and “true” heteroskedasticity as the two problems call for different remedial action. Thursby 1982 shows that the combined use of the RESET and Goldfeld-Quandt provide a way to distinguish the two.⁵ Early work by Zarembka 1974 suggested that, since data transformations may stabilize the variance or induce heteroskedasticity, the Box-Cox procedure is not robust to heteroskedastic errors, biasing results toward variance-stabilizing transformations. Lahiri and Egy 1981 showed that, if z is a variable related to the pattern of heteroskedasticity according to $\sigma_i^2 = \sigma^2(z_i^\delta)$, δ and θ can be jointly estimated, and separate transformations for misspecification and heteroskedasticity can be implemented.

3.3 Collinearity

A common feature of hedonic regression is a large number of explanatory variables. Since most hedonic research uses observational data, some degree of collinearity is unavoidable, causing potentially noisy estimates. If the implicit prices are of minor interest per se, as when building hedonic price indices, collinearity is of limited concern. If instead valuing product attributes is the primary objective of the analysis, remedial action should be considered. The fact that coefficients in hedonic regression can often be bounded, at a minimum in term of their signs, inspired the Bayesian approach (e.g., Atkinson and Crocker 1987; Gilley and Pace 1995), yet non-frequentist methods have seen limited application so far in the hedonic literature.

A more common approach is reducing the dimensionality of the data matrix with the use of multivariate analysis techniques such as principal components (Asher 1992), or factor analysis (Rosiers, Thériault, and Villeneuve 2000). One caveat associated with the adoption of such techniques is that the resulting data constructs may lack a clear economic interpretation, which defeats the purpose of the modeling exercise when the primary objective is a study of implicit price functions. If a particular subset of coefficients is of interest, a possible approach is applying dimensionality-reducing techniques to all variables but the ones of direct interest.

3.4 Specification Search Example: A Hedonic Price Equation for Wine

An example of a semiparametric model of a hedonic price function for red wine is offered in this section. This example is based on a standard specification search performed in Costanigro, McCluskey, and Mittelhammer (2007). The section then proceeds to show price-dependent market segmentation in the wine market. The data set is composed of 13,024 observations derived from ten years (1991–2000) of ratings scores reported in the *Wine Spectator* magazine for California and Washington red wines. The variables include the price of the wine adjusted to 2000 values by a consumer price index (CPI) for alcohol, score obtained in expert sensory evaluation by the *Wine Spectator's* experts, the number of cases produced, and the years of aging before commercialization. Descriptive statistics for these variables are reported in Table 6.1. Indicator variables were used to denote regions of production, wine varieties, and the presence of label information. Table 6.2 presents the variables with a brief description.

A series of possible transformations of the dependent variable were considered and evaluated on the basis of variance stabilization, normality of the residuals, and misspecification. The Goldfeld-Quandt test was used to detect heteroskedasticity proportional to predicted values and the RESET test for misspecification (second, third, and fourth power). For the normality of the residuals, we employed three different tests: Anderson-Darling, Komolgorov-Smirnov and Ryan-Joiner. Results of the tests are presented in Table 6.3. The inverse square transformation performed the best. The specifications of the independent variables were also determined by screening possible transformations of the non-binary variables and examining excluded-variable residual plots. The following hedonic price function was estimated via OLS:

Hedonic Price Analysis in Food Markets

Table 6.1 Descriptive statistics of quantitative explanatory variables

State	variable	Mean	Median	Minimum	Maximum
California ^a	Price ^b	31.1	22	3	2,000
	Cases	6,719	1,467	16	950,000
	Score	86.1	87	60	99
	Age	2.8	3	1	9
Washington ^c	Price ^b	23.3	20	5	144
	Cases	6,720	1,000	45	550,000
	Score	86.8	87	67	96
	Age	2.8	3	1	7

(a) 11,774 observations.

(b) Adjusted to year 2000 by a CPI index for alcohol.

(c) 1,250 observations.

Table 6.2 Descriptions of the abbreviation used for the explanatory variables (11)

Predictor	short description
Score	Rating score from the wine spectator
Scscore	Score centered by subtracting its mean
Scscore ²	Scscore squared
Age	Years of aging before commercialization
Agesc	Age centered by subtracting its mean
Agesc ²	Agesc squared
Cases	Number of cases produced
Lncas	Natural log of hundreds of cases produced
Napa	Region of production
Bay area	
Sonoma	
South coast	
Carneros	
Sierra foothills	
Mendocino	
Washington	
Non-varietal	Grape variety
Pinot Noir	
Cabernet	
Merlot	
Syrah	
Reserve	"Reserve" was reported on the label
Vineyard	Specific name of the vineyard on the label
Estate	"Estate" produced wine
91, ..., 99	Vintage
Wa	Washington state wines

$$\begin{aligned} Price_i^{-0.5} = & \beta_0 + \beta_0^w + (\beta_1)(score_i) + (\beta_2)(score_i)^2 + (\beta_3)(Age_i) + (\beta_4)(Age_i)^2 \\ & + (\beta_5)LN(cases_i) + \sum_{k=1}^5 (\beta_5 + k)(Variety_i) + \sum_{k=1}^9 (\beta_{10} + k)(Vintage_i) \\ & + \sum_{k=1}^3 (\beta_{19} + k)(Label_i) + \sum_{k=1}^7 (\beta_{22} + k)(Region_i) + \varepsilon_i \end{aligned}$$

. Variables interacting Washington state with other attributes were also estimated but are not reported owing to their statistical insignificance. As Table 6.3 shows, formal testing still detected a moderate degree of heteroskedasticity (and misspecification),⁶ but the possible gains in estimation efficiency that might be achieved by adjusting the estimator for an appropriate heteroskedastic process are muted by the consistency of

Hedonic Price Analysis in Food Markets

Table 6.3 Test statistics result for several specifications of the dependent variable

Transformation of dependent variable	Normality			Specification			Heteroskedasticity
	Anderson-A-squared	Ryan-Joiner R	Komolgorov-Smirnov D	Reset (2) F-value	Reset (3) F-value	Reset (4) F-value	Goldfeld-Quandt GQ
-2	552.548	0.879	0.155	1237.100	692.730	469.090	20.497
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-1.5	295.709	0.938	0.114	713.450	436.320	294.860	9.029
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-1	106.373	0.979	0.066	228.520	172.030	116.840	3.923
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
-0.5	18.584	0.997	0.025	1.236	26.907	19.097	1.618
	0.000	0.000	0.000	0.266	0.000	0.000	0.000
Box-Cox (-0.36)	14.972	0.997	0.023	9.3751	23.5	***	1.2214
	0.000	0.000	0.000	0.002	0.000	0.000	0.000
Natural log	62.396	0.974	0.048	187.000	101.960	69.217	1.807
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.5	581.245	0.804	0.140	543.330	276.400	184.920	8.911
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Linear	2,200.000	0.490	0.285	465.850	242.590	168.760	92.490
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1.5	3,400.000	0.305	0.385	264.230	147.330	101.640	777.130
	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2	3,800.000	0.233	0.421	147.590	82.081	59.130	12,204.150
	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: Probability values are displayed below statistical test values.

the OLS estimator and the large sample size on which the estimators are based.⁷ As a matter of caution the covariance matrix of the parameters was estimated using White's consistent heteroskedasticity-robust estimator.

Estimated coefficients of the model in (11) are reported in Table 6.4. The large tstatistics suggest that collinearity does not represent an issue in this sample. In interpreting the results, it is important to note that owing to the transformation of

Table 6.4 Hedonic price estimation for red wine

Variable	coefficient	t
Constant	21.999	104.7

Hedonic Price Analysis in Food Markets

Scscore	-0.620	-61.29
Scscore ²	-0.022	-16.47
Agesc	-1.302	-23.69
Agesc ²	0.108	2.63
Lncas	1.004	41.75
Napa ^a	-5.483	-36.77
Bay area ^a	-3.437	-17.34
Sonoma ^a	-4.053	-28.31
South coast ^a	-3.222	-20.46
Carneros ^a	-4.291	-23.74
Sierra foothills ^a	-2.327	-10.3
Mendocino ^a	-2.406	-12.64
Wa ^a	-0.652	-0.95
Non-varietal ^b	-4.319	-27.29
Pinot Noir ^b	-3.252	-32.16
Cabernet ^b	-2.479	-24.4
Merlot ^b	-2.200	-21.79
Syrah ^b	-0.582	-4.26
Reserve ^c	-1.105	-10.89
Vineyard ^c	-0.858	-11.46
Estate ^c	-0.601	-2.88
91 ^d	5.353	31.78
92 ^d	5.339	1.38
93 ^d	4.372	27.24
94 ^d	4.097	27.8
95 ^d	3.311	23.24
96 ^d	2.397	17.75
97 ^d	1.749	13.07
98 ^d	0.393	2.77
99 ^d	0.668	5.02

(a) Omitted dummy variable: generic California.

(b) Omitted dummy variable: Zinfandel.

(c) Omitted dummy variable: no additional label information.

(d) Omitted variable: year 2000.

the dependent variable, coefficients with a negative sign signify a

positive marginal implicit price. Empirical results conformed to a priori expectations: for all estimated models, price is increasing in aging and rating score over the range of the data and decreasing in the number of cases produced. Confirming previously published results, regional appellations command price premiums relative to a generic California wine, with Napa Valley bringing the largest premium. The coefficients associated with the variety variables capture the difference in price relative to Zinfandel grapes, and the coefficients for vintages refer to price differences relative to the excluded year 2000. This empirical example was a first step in understanding the Washington and California red wine markets.

3.5 Some Considerations on the Retransformation Problem

In hedonic modeling, non-linear transformations of the dependent variable induce implicit price functions that depend on the price of the differentiated product, which in turn is a function of the levels of all attributes. If we consider a deterministic model in the form $h(p) = z\beta$, where $h(p)$ represents a non-linear transformation, the implicit price functions can be recovered as $\frac{\partial p}{\partial z_k} = \frac{\partial[h^{-1}(z\beta)]}{\partial z_k} = \frac{\partial h^{-1}(p)}{\partial p} \frac{\partial z\beta}{\partial z_k}$, where $\frac{\partial h^{-1}(p)}{\partial p}$ is a function of p in virtue of the non-linearity of h . In many applications, this is a desirable feature. For example, when estimating a hedonic model of wine prices, we might expect the implicit price of one extra year of cellaring to be proportional to the total price of the product. One down side is that once a transformation of the dependent variable is adopted, it is applied indiscriminately to the implicit price function of all attributes. An additional problem is that the original units of measure are lost, and applied economists and policymakers alike are interested in the dollar value of goods and product attributes, not log dollars or inverse square root dollars. To restore the original units of measure, inverse transformations (e.g., exponentiation after the natural log) are often applied⁸ on predicted values and estimated coefficients to restore the original units of measure. While this is legitimate within the deterministic model presented before, Jensen's inequality implies that generally⁹ $E(y|x) \neq h^{-1}(E(h(y)|x))$ and $\frac{\partial E(y|x)}{\partial x} \neq \frac{\partial h^{-1}[E(h(y)|x)]}{\partial x}$.

The issue of retransformation has been largely ignored in hedonic modeling, but has been analyzed within the context of log-linearization of multiplicative models (e.g., Cobb-Douglas in production models and gravity equations in international trade), but the implications straightforwardly extend to hedonic models. While early results by Goldberger 1968 on log-normal distributions suggested that only estimates of the intercept are affected by retransformation bias, which could be easily eliminated, more recent work by Silva and Tenreiro (2006: 664) show that, in the presence of an unknown form of heteroskedasticity, "the non-linear transformation of the dependent variable changes the properties of the error term in a non-trivial way" and therefore "it is not possible to recover information about the conditional expectation of y from the conditional mean of $\ln(y)$."

Manning 1998 presents a broader treatment of the issue within the context of models of health care expenditure, where, much like in hedonics, non-linear transformations are implemented to deal with a heavily right-skewed dependent variable. Modifying the notation to highlight our application to hedonic modeling, the author first considers the familiar log-linear model $\ln(p) = z\beta + \varepsilon$; with $E(\varepsilon) = 0$, $E(\varepsilon^2) = \sigma^2$, and non-constant variance $E(\varepsilon^2) \neq c$ to show that: (12)

$$E(p/z) = e^{z\beta} E(e^\varepsilon) = e^{z\beta} \int e^\varepsilon dF(\varepsilon) \neq e^{z\beta}$$

where $F(\varepsilon)$ is the cumulative distribution of ε . If the assumption that ε is distributed $N(0, \sigma^2(Z))$ is made, implying that prices are log-normally distributed but the variance is proportional to one or more of the explanatory variables, the conditional expectation of the original variable can be recovered as (13)

$$E(p/z) = e^{z\beta + 0.5\sigma^2(z)}$$

and (14)

$$\frac{\partial E(p/z_k)}{\partial z_k} = E(p/z) \left[\beta_k + 0.5 \frac{\partial \sigma^2(z)}{\partial z_k} \right]$$

Substitution of $\sigma^2(Z)$ with σ^2 in equations (13) and (14) produces Goldberger's (1968) results regarding bias correction adjustment and unbiasedness of retransformed estimates of the slope parameters under homoskedasticity. For the general case of a BoxCox

transformation and unknown price distribution to recover $E(p|z)$ the integral $\int [1 + \theta(z\beta + \varepsilon)]^{\frac{1}{\theta}} dF(\varepsilon)$ needs to be evaluated. In

Manning's words, "The price of using a log (or any other) transform of the dependent variable is that the analyst must also learn about the nature of the error structure" (1998: 293). A possible approach to avoid the retransformation problem is estimating the hedonic price function at multiple conditional quantiles (Koenker and Basset 1978), leaving the dependent variable in its linear form. In this way,

implicit prices can be immediately recovered from the estimated coefficient, and are allowed to vary across (predicted) prices. This method has been used in Costanigro, McCluskey, and Goemans (2010), but its general applicability to multiple hedonic settings is the object of ongoing work.

3.6 Non-Parametric and Hybrid Models

Consider the following econometric model: (15)

$$p = g(z) + \varepsilon,$$

where no assumptions are made regarding the probability density function of ε , and $g(z)$ is left unspecified. The appealing feature of the hedonic model in (15) is that the analyst only needs to identify the product attributes appearing in Z , leaving the determination of the functional dependency of prices and attributes to be completely data-driven.

Kernel regression estimators, of which Nadaraya-Watson is a popular example, are able to estimate $E(p|z)$ in this “assumption-free” environment by evaluating $\hat{g}(z)$ in a series of piecewise estimation using only data in the proximity (i.e., with similar values in the covariates) of a given point of evaluation (p_0, z_0). The general idea¹⁰ is that, if Kernel density estimators can be used to estimate the joint and marginal density functions $f(p, z)$ and $f(z)$, then one can reconstruct the shape of the conditional density function

$$E(p/z) = g(z) = \int_{-\infty}^{\infty} p \left[\frac{f(p, z)}{f(z)} \right] dp.$$

For each point of evaluation, a non-parametric estimate of $g(z_0)$ can be represented as a weighted function of the dependent variable: (16)

$$\hat{P}_0 = \hat{g}(z_0) = \sum_{i=1}^N w_i(z_0; z_i; h) p_i,$$

where $w_i(z_0; z_i; h)$ emphasizes that the weight assigned by the Kernel function to each observation is proportional to some measure¹¹ of distance between each set of covariates in the pairs $(z_{0k}; z_{ik})$. The parameter h , named bandwidth, determines the size of the data neighborhoods, and is either set by the researcher or determined by some pre-or post-data optimality criterion. For each point of evaluation, data points falling outside of the neighborhood receive a zero weight. The whole surface of the conditional density is reconstructed by estimating $\hat{g}(z)$ at multiple points of evaluation, thereby accommodating for virtually any functional relationship between regressand and regressors.

The cost of eliminating all structure and assumptions from the econometric model is that non-parametric estimates tend to be noisy and imprecise. In the hedonic context, the problem is exacerbated by the large number of regressors and the so-called curse of dimensionality (Bellman 1961): for a given point of evaluation ($p_0; z_0$) and a fixed choice of the bandwidth parameter h , increasing the number of regressors reduces the number of observations close to a point of evaluation, making the data sparse and the estimates imprecise. Thus, large and “dense” data sets are necessary to achieve acceptably precise estimates.

While Nadaraya-Watson-type estimators have seen few applications in the hedonic literature, applied economists have used a number of strategies to conjugate the appealing features of non-parametric methods and the necessity of introducing some structure in the econometric model. One approach is the joint use of variable bandwidth algorithms that include more data when the design matrix becomes sparse, such as the locally weighted scatterplot smoothing (LOESS) (Cleveland and Devlin 1988), and the adoption of polynomial expansions accommodating for the possible curvature of $g(z)$ when a sparse design matrix induces larger—potentially “not local”—neighborhoods of data (see Trevor Hastie and Loader 1993). For the case of a linear (first-order) approximation to $g(z_0)$, the local polynomial estimator (Fan and Gijbels 1996: 58) solves the minimization problem: (17)

$$\arg \min_{\hat{\alpha}, \hat{\beta}} \left\{ \sum_{i=1}^n [p_i - \hat{\alpha}_{(z_0)} - (z_i - z_0)\hat{\beta}_{(z_0)}]^2 K[(z_i - z_0)/h] \right\}$$

where K represents a Kernel weighting function of choice.¹² It should be noted that Nadaraya-Watson-type estimators can be conceptualized as zero-order polynomial expansions, in which only the intercept $\hat{\alpha}_{(z_0)}$ in (17) is estimated. In addition, to allow for some curvature, the local linear regressor of (17) makes implicit prices immediately available, as the estimate of $\frac{\partial E(p/z_0)}{\partial z_k}$ can be found in the k th entry of $\hat{\beta}_{(z_0)}$. Estimation is implemented via a series of weighted OLS regressions; one at each point of evaluation, where the weights are determined by the Kernel function and each row of z is transformed according to $z_i^* = [z_i - z_0]$. Applications of Local Polynomial Regression to hedonic modeling are becoming increasingly common, and include Bin 2004, Martins-Filho and Bin 2005, Bajari and Benkard 2005, Bontemps, Simioni, and Surry (2008), and Costanigro, Mittelhammer, and McCluskey (2009).

Another way of imposing some structure to the econometric model is estimating hybrid functions (e.g., Anglin and Gencay 1996; Bontemps, Simioni, and Surry 2008) in which a set of attributes (x) enter the econometric model parametrically while other attributes (z) are estimated non-parametrically, thereby obtaining the econometric model (18)

$$p = x\beta + g(z) + \varepsilon;$$

where x may include a transformed variable. Qualitative regressors, which complicate non-parametric estimation,¹³ and control variables of minor economic interest are generally included in the parametric part, while the non-parametric approach is reserved for the subset of variables of principal economic interest.

The back-fitting approach (Friedman and Stuetzle 1981; Hastie and Tibshirani 1990) adds even more structure to the model, estimating hybrid models in which the functional form of a set of variables is left unspecified, but an assumption of additivity is made: (19)

$$p = x\beta + \sum_{j=1}^j g_j(z_j) + \varepsilon$$

Estimation of each $g_j(z_j)$ is accomplished iteratively in a series of non-parametric regressions on excluded-variable residuals (see Bin 2004 and Martins-Filho and Bin 2005 for an application to hedonic models).

Because of their computationally intensive nature, non-parametric methods were rarely employed in the early years of the hedonic literature, but are now becoming increasingly common in the face of their generally superior predictive performance and the increase in computing power available to applied economists. Recent theoretical contributions in hedonic modeling (Ekeland, Heckman, and Nesheim 2002; Bajari and Benkard 2005) show that the non-linearity of implicit price functions can also serve as a base for the identification of the second stage of hedonic regression in single markets data suggest that the trend will continue.

3.7 Aggregation and Market Segmentation

Since the hedonic model assumes the market is in equilibrium and there is perfect information, products with similar vectors of attributes should share similar implicit price functions. If price differences arise, perfectly informed buyers will purchase only the cheaper products, until the equilibrium is re-established. Therefore, diverging implicit price functions, when encountered, provide some evidence for the existence of market segments.

The debate over the relevance of market segmentation in hedonic modeling originated in the real estate literature with Straszheim 1974, who noted that implicit price functions changed across San Francisco districts and segmenting the hedonic regression accordingly increased the overall explanatory power of the hedonic model. When submarkets exist, he noted, estimating a single aggregate hedonic function will generally produce biased estimates. According to Freeman 1993: 368), a fundamental condition for the existence of different price functions is that "the structure of demand, the structure of supply, or both, must differ across segments." This stems from the fact that the hedonic price function is a double envelope of bid (demand) and offer (supply) functions. Furthermore, some form of obstacle to mobility must impede sellers from arbitraging and buyers from migrating to cheaper markets.

While segmenting factors in markets for differentiated products include the three major dimensions of space, time, and product characteristics, the literature on market segmentation largely focused on geographical and spatial factors, owing to the original context of modeling real estate prices. To provide a few examples within this vast stream of literature, Straszheim 1974 considered political boundaries, Goodman and Thibodeau 1998 differences in school district performance and ethnic composition across districts. Along similar lines, the work of environmental economists quantifying how vicinity to amenities (e.g., Tyrväinen and Miettinen 2000) or nuisance factors (e.g., Bontemps, Simioni, and Surry 2008) influences property values is also a (sometimes implicit) exercise in identifying submarkets. In food markets, spatial market segmentation often arises from differences in consumer preferences across countries (consider, for example, the different valuation of the "organic" attribute across countries observed by Baker et al. 2004).

Temporal differences in equilibrium prices are at the core of the hedonic price indexing literature (see Triplett 2006 for a general survey), where the hedonic framework is used to measure changes in prices across time periods holding constant product attributes. In other hedonic studies, price indices are routinely used to control for such temporal variations.

Market segmentation induced by product characteristics generally arises from the bundled nature of differentiated products. One possibility is that certain combinations of attributes may not be produced or available for purchase. If the attribute space is not a convex set, researchers may run the risk of extrapolating their results. More importantly, Pakes 2003 notes that, when (proprietary) attributes exist as a result of product innovation, market power will distort the implicit price function of all the attributes bundled with the improved product.

Costanigro, Mittelhammer, and McCluskey (2009) consider use-based market segments, noting that

as the vectors of attribute levels characterizing two goods increasingly diverge, it is more likely that consumers purchase the goods for different purposes. That is, the more two products are differentiated, the less they are fungible in the same or similar use. Further, the costs associated with assembling a given bundle of attributes in the same product will change as the vector of attributes changes.

Even though their application is the wine market, the concept applies more generally. Consider the automotive industry: SUVs and sports cars are both vehicles, but they are generally used for different purposes. Many product attributes may be similar across these types of vehicle, yet the structure of the buyers' preferences, the number of competing firms, and their cost structure will change.

Once a reason for suspecting market segmentation is identified, an empirical strategy to ultimately estimate separate hedonic functions is needed. The most immediate approach is to isolate a number of data partitions (i.e., the market segments) for which a unique hedonic function can be estimated. In some cases the presence of one or more product attributes unambiguously identifies a market

segment, and the categorization process is rather trivial (e.g., estimate separate regressions for offices, apartments, and detached houses).

When the transition from one market segment to another is gradual and a clear-cut rule for isolating market segments does not exist, the empirical task is more challenging. Such complications often arise when multiple concurring factors generate the market segments, or when a gradual transition in segmenting factors generates partially overlapping market segments. One approach is to apply data clustering algorithms (Ward and K-means are frequent choices) to the original data or to factor scores and principal components (e.g., Bourassa et al. 1999; Watkins 1999), thereby identifying market segments on the basis of similarities in attribute levels. For this method to produce desirable results, the researcher needs to know which attributes to include in the clustering step (i.e., the segmenting factors), and a direct, constant (i.e., no threshold or qualitative effects) relationship between changes in attribute levels and changes in implicit price functions needs to hold. Costanigro, Mittelhammer, and McCluskey (2009) pointed out that these conditions rarely hold, and proposed a multi-step process in which the hedonic function is first estimated locally (i.e., non-parametrically) and then using a clustering algorithm to isolate data partitions with similar implicit price functions. The remaining drawback, common to all clustering approaches, is that the classes of products identified by each data partition may not be immediately clear to the researcher.

A second approach is making market segmentation an explicit part of the hedonic model. In certain cases, segmenting factors can be included as extra product attributes (e.g., positive and negative externalities in environmental studies), and varying implicit price functions can be accommodated via appropriate interaction terms between attributes and segmenting factors. When more than one segmenting factor exists, or segmenting factors are not known to the researcher, latent class (Aitkin and Rubin 1985) and mixture models (Heckman and Singer 1984) are appropriate, even though these methods have seen little application in the hedonic context. The principal drawback of these likelihood-based techniques is their heavy parameterization: the researcher needs to commit to a density function describing the price distribution, and one describing the genesis of submarkets.

4 Conclusions

Increasingly food products are heterogeneous and have multiple characteristics. This is a relatively recent phenomenon and is in contrast to the food commodities that were purchased by previous generations of consumers. The mass customization of purchases changes how researchers can most effectively study food markets. Work is needed to determine how to analyze, both conceptually and empirically, the increased differentiation and variety that exists in today's food markets.

Many interesting new topics of research are emerging in this area of inquiry. Not only will agricultural economists need to combine information about consumer preferences that are present in different sources of data, they will also need to turn to other disciplines to better understand markets. This includes insights from sensory scientists about factors that affect taste and eating quality, and concepts from marketing and psychology to investigate how values, culture, identity, and social norms affect markets.

From an empirical point of view, the increasing globalization of the world food markets offers the opportunity to use spatial statistical techniques to investigate to what extent multiple markets overlap and interact in determining food prices. The use of spatial information helps to control for omitted variables correlated with space. Such variables in food markets could include location of production origin (especially important for products that rely on collective reputations such as wine), preferences for organic foods and other socially responsible food attributes, and the general availability of variety.

In this chapter we discussed theoretical and empirical issues related to both hedonic theory and product characteristics models. There are many empirical issues to address and potential problems to overcome with both approaches. This is a fruitful area of research with increased customization of products and the increasing availability of large data sets, including scanner data from the food industry.

References

- Adelman, I., and Z. Griliches. 1961. "On an Index of Quality Change." *Journal of the American Statistical Association* 56: 535–48.
- Aitkin, M., and D. B. Rubin. 1985. "Estimation and Hypothesis Testing in Finite Mixture Models." *Journal of the Royal Statistical Society, ser. B: Methodological* 47: 67–75.
- Aldy, J. E., and W. K. Viscusi. 2008. "Adjusting the Value of a Statistical Life for Age and Cohort Effects." *Review of Economics and Statistics* 90/3: 573–81.
- Anderson, S. 2008. "Product Differentiation." In S. N. Durlauf and L. E. Blume, eds, *The New Palgrave Dictionary of Economics*. Basingstoke: Palgrave Macmillan.
- Anglin, P. M., and R. Gençay. 1996. "Semiparametric Estimation of a Hedonic Price Function." *Journal of Applied Econometrics* 11: 633–48.
- Asher, C. C. 1992. "Hedonic Analysis of Reliability and Safety for New Automobiles." *Journal of Consumer Affairs* 26/2: 377.
- Atkinson, S. E., and T. D. Crocker. 1987. "A Bayesian Approach to Assessing the Robustness of Hedonic Property Value Studies." *Journal of Applied Econometrics* 2/1: 27–45.
- Bajari, P., and C. L. Benkard. 2005. "Demand Estimation with Heterogeneous Consumers and Unobserved Product Characteristics: A

Hedonic Price Analysis in Food Markets

Hedonic Approach." *Journal of Political Economy* 113/6: 1239–76.

Baker, S., K. E. Thompson, J. Engelken, and K. Huntley. 2004. "Mapping the Values Driving Organic Food Choice: Germany versus the UK." *European Journal of Marketing* 38/8: 995–1012.

Barkema, A., and M. Drabenstott. 1995. "The Many Paths of Vertical Coordination: Structural Implications for the U.S. Food System." *Agribusiness* 11: 483–92.

Bartik, T. J. 1987. "The Estimation of Demand Parameters in Hedonic Price Models." *Journal of Political Economy* 95/1 (Feb.), 81–8.

Bellman, R. 1961. *Adaptive Control Processes: A Guided Tour*. Princeton: Princeton University Press.

Berry, S. T., J. Levinsohn, and A. Pakes. 1995. "Automobile Prices in Market Equilibrium." *Econometrica* 64: 841–90.

Bin, O. 2004. "A Prediction Comparison of Housing Sales Prices by Parametric versus SemiParametric Regressions." *Journal of Housing Economics* 13: 68–84.

Bontemps, C., M. Simioni, and Y. Surry. 2008. "Semiparametric Hedonic Price Models: Assessing the Effects of Agricultural Nonpoint Source Pollution." *Journal of Applied Econometrics* 23/6: 825–42.

Bourassa, S. C., F. Hamelink, M. Hoesli, and B. D. MacGregor. 1999. "Defining Housing Submarkets." *Journal of Housing Economics* 8: 160–83.

Box, G. E. P., and D. R. Cox. 1964. "An Analysis of Transformations." *Journal of the Royal Statistical Society, ser. B: Methodological* 26/2: 211–52.

Brown, J. N., and H. S. Rosen. 1982. "On the Estimation of Structural Hedonic Price Models." *Econometrica* 50/3: 765–8.

Carew, R. 2000. "A Hedonic Analysis of Apple Prices and Product Quality Characteristics in British Columbia." *Canadian Journal of Agricultural Economics* 48: 241–57.

— and W. J. Florkowski. 2010. "The Importance of Geographic Wine Appellations: Hedonic Pricing of Burgundy Wines in the British Columbia Wine Market." *Canadian Journal of Agricultural Economics* 58/1: 93–108.

Cassel, E., and R. Mendelsohn. 1985. "The Choice of Functional Forms for Hedonic Price Equations: Comment." *Journal of Urban Economics* 18/2: 135–42.

Christensen, L. R., D. W. Jorgenson, and L. J. Lau. 1973. "Transcendental Logarithmic Production Frontiers." *Review of Economics and Statistics* 55/1: 28–45.

Cleveland, W. S., S. J. Delvin, and E. Grosse. 1988. "Regression by Local Fitting." *Journal of Econometrics* 37: 87–114.

Combris, P., S. Lecocq, and M. Visser. 1997. "Estimation of a Hedonic Price Equation for Bordeaux Wine: Does Quality Matter?" *Economic Journal* 107: 390–403.

Costanigro, M., J. McCluskey, and C. Goemans. 2010. "The Economics of Nested Names: Name Specificity, Reputation and Price Premia." *American Journal of Agricultural Economics* 92/5 (Oct. 1), 1339–50.

—, and R. C. Mittelhammer. 2007. "Segmenting the Wine Market Based on Price: Hedonic Regression When Different Prices Mean Different Products." *Journal of Agricultural Economics* 58/3: 454–66.

—, R. C. Mittelhammer, and J. J. McCluskey. 2009. "Let the Market Be Your Guide: Estimating Equilibria in Differentiated Product Markets with Class-Membership Uncertainty." *Journal of Applied Econometrics* 24: 1117–35.

Court, A. T. 1939. "Hedonic Price Indexes with Automotive Examples." In *The Dynamics of Automobile Demand*. New York: General Motors.

Court, L. M. 1941. "Entrepreneurial and Consumer Demand Theories for Commodities Spectra." *Econometrica* 9: 135–62.

Durham, C. A., I. Pardoe, and E. Vega. 2004. "A Methodology for Evaluating How Product Characteristics Impact Choice in Retail Settings with Many Zero Observations: An Application to Restaurant Wine Purchase." *Journal of Agricultural and Resource Economics* 29/1: 112–31.

Ekeland, I., J. J. Heckman, and L. Nesheim. 2002. "Identifying Hedonic Models." *American Economic Review* 92/2: 304–9.

———. 2004. "Identification and Estimation of Hedonic Models." *Journal of Political Economy* 112/S1: S60–S109.

Epplé, D. 1987. "Hedonic Prices and Implicit Markets: Estimating Demand and Supply Functions for Differentiated Products." *Journal of Political Economy* 95/1: 59–80.

Espinosa, J. A., and B. K. Goodwin. 1991. "Hedonic Price Estimation for Kansas Wheat Characteristics." *Western Journal of Agricultural Economics* 16: 72–85.

Hedonic Price Analysis in Food Markets

- Fan, J., and I. Gijbels. 1996. *Local Polynomial Modelling and its Applications*, 1st edn. London: Chapman and Hall/CRC, Mar. 1.
- Faux, J., and G. M. Perry. 1999. "Estimating Irrigation Water Value Using Hedonic Price Analysis: A Case Study in Malheur County, Oregon." *Land Economics* 75/3: 440–52.
- Fletcher, M., J. Mangan, and E. Raeburn. 2004. "Comparing Hedonic Models for Estimating and Forecasting House Prices." *Property Management* 22/3: 189–200.
- Freeman, A. M. 1993. *The Measurement of Environmental and Resource Values: Theory and Methods*. Washington, DC: Resources for the Future.
- Friedman, J. H., and W. Stuetzle. 1981. "Projection Pursuit Regression." *Journal of the American Statistical Association* 76: 817.
- Gilley, O. W., and R. K. Pace. 1995. "Improving Hedonic Estimation with an Inequality Restricted." *Review of Economics and Statistics* 77/4: 609.
- Goldberger, A. S. 1968. "The Interpretation and Estimation of Cobb-Douglas Functions." *Econometrica* 36/3–4: 464–72.
- Goodman, A. C., and T. G. Thibodeau. 1998. "Dwelling Age Heteroskedasticity in Repeat Sales House Price Equations." *Real Estate Economics* 26/1: 151–71.
- Griliches, Z., ed. 1971. *Price Indexes and Quality Change*. Cambridge, MA: Harvard University Press.
- Halvorsen, R., and H. O. Pollakowski. 1981. "Choice of Functional Form for Hedonic Price Equations." *Journal of Urban Economics* 10/1: 37–49.
- Hastie, T., and C. Loader. 1993. "Local Regression: Automatic Kernel Carpentry: Rejoinder." *Statistical Science* 8/2: 139–43.
- and R. J. Tibshirani. 1990. *Generalized Additive Regression*. London: Chapman Hall.
- Heckman, J., and B. Singer. 1984. "A Method for Minimizing the Impact of Distributional Assumptions in Econometric Models for Duration Data." *Econometrica* 52/2: 271–320.
- Huang, C. L., and B.-H. Lin. 2007. "A Hedonic Analysis of Fresh Tomato Prices among Regional Markets." *Review of Agricultural Economics* 29/4: 783–800.
- Koenker, R. 2005. *Quantile Regression*. Cambridge: Cambridge University Press.
- and G. Bassett. 1978. "Regression Quantiles." *Econometrica*, 46/1 (Jan.), 33–50.
- Lahiri, K., and D. Ego. 1981. "Joint Estimation and Testing for Functional Form and Heteroskedasticity." *Journal of Econometrics* 15/2: 299–307.
- Lancaster, K. 1966. "A New Approach to Consumer Theory." *Journal of Political Economy* 74: 132–57.
- Landon, S., and C. E. Smith. 1997. "The Use of Quality and Reputation Indicators by the Consumers: The Case of Bordeaux Wine." *Journal of Consumer Policy* 20: 289–323.
- . 1998. "Quality Expectations, Reputation, and Price." *Southern Economic Journal* 64/3: 628–47.
- Loureiro, M. L., and J. J. McCluskey. 2000. "Assessing Consumer Response to Protected Geographical Identification Labeling." *Agribusiness* 16/3: 309–20.
- McCluskey, J. J., and G. C. Rausser. 2003. "Stigmatized Asset Value: Is It Temporary or Long-Term?" *Review of Economics and Statistics* 85/2: 276–85.
- McConnell, K. E., and I. E. Strand. 2000. "Hedonic Prices for Fish: Tuna Prices in Hawaii." *American Journal of Agricultural Economics* 82/1: 133–44.
- Manning, W. G. 1998. "The Logged Dependent Variable, Heteroskedasticity, and the Retransformation Problem." *Journal of Health Economics* 17/3: 283–95.
- Martins-Filho, C., and O. Bin. 2005. "Estimation of Hedonic Price Functions via Additive Non-parametric Regression." *Empirical Economics* 30/1: 93–114.
- Mittelhammer, R. C., G. G. Judge, and D. J. Miller. 2000. *Econometric Foundations*. Cambridge: Cambridge University Press.
- Nerlove, M. 1995. "Hedonic Price Functions and the Measurement of Preferences: The Case of Swedish Wine Consumers." *European Economic Review* 39: 1697–1716.
- Oczkowski, E. 1994. "Hedonic Wine Price Function for Australian Premium Table Wine." *Australian Journal of Agricultural Economics* 38: 93–110.

- Pakes, A. 2003. "A Reconsideration of Hedonic Price Indexes with an Application to PCs." *American Economic Review* 93/5: 1578–96.
- Palmquist, R. B. 1984. "Estimating the Demand for the Characteristics of Housing." *Review of Economics and Statistics* 66/3 (Aug.), 394–404.
- Parcell, J. L., and T. C. Schroeder. 2007. "Hedonic Retail Beef and Pork Product Prices." *Journal of Agricultural and Applied Economics* 39: 29–46.
- Parmeter, C. F., D. J. Henderson, and S. C. Kumbhakar. 2007. "Non-parametric Estimation of a Hedonic Price Function." *Journal of Applied Econometrics* 22/3: 695–9.
- Quigley, J. M. 1982. "Non-linear Budget Constraints and Consumer Demand: An Application to Public Programs for Residential Housing." *Journal of Urban Economics* 12/2 (Sept.), 177–201.
- Racine, J., and Q. Li. 2004. "Non-parametric Estimation of Regression Functions with both Categorical and Continuous Data." *Journal of Econometrics* 119/1: 99–130.
- Ramsey, J. B. 1969. "Tests for Specification Errors in Classical Linear Least-Squares Regression Analysis." *Journal of the Royal Statistical Society, ser. B: Methodological* 31/2: 350–71.
- Rosen, S. 1974. "Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition." *Journal of Political Economy* 82: 34–55.
- 1981. "The Economics of Superstars." *American Economic Review* 71/5 (Dec.), 845–58.
- Rosiers, F. D., M. Thériault, and P. Villeneuve. 2000. "Sorting Out Access and Neighbourhood Factors in Hedonic Price Modelling." *Journal of Property Investment and Finance* 18/3: 291–315.
- Santos Silva, J. M. C., and S. Tenreyro. 2006. "The Log of Gravity." *Review of Economics and Statistics* 88/4: 641–58.
- Schamel, G., and K. Anderson. 2003. "Wine Quality and Varietal, Regional and Winery Reputations: Hedonic Prices for Australia and New Zealand." *Economic Record* 79: 357–69.
- Stanley, L. R., and J. Tschirhart. 1991. "Hedonic Prices for a Non-durable Good: The Case of Breakfast Cereals." *Review of Economics and Statistics* 73/3: 537–41.
- Steiner, B. 2004. "French Wines on the Decline? Econometric Evidence from Britain." *Journal of Agricultural Economics* 55/2: 267–88.
- Straszheim, M. 1974. "Hedonic Estimation of Housing Market Prices: A Further Comment." *Review of Economics and Statistics* 56/3: 404–6.
- Taylor, L. O. 2003. "The Hedonic Method." In P. A. Champ, K. J. Boyle, and T. C. Brown, eds, *A Primer on Non-Market Valuation*. Boston: Kluwer Academic Publishers.
- Thursby, J. G. 1982. "Misspecification, Heteroscedasticity, and the Chow and Goldfeld-Quandt Tests." *Review of Economics and Statistics* 64/2: 314–21.
- Triplett, J. 2006. *Handbook on Hedonic Indexes and Quality Adjustments in Price Indexes*. Paris: OECD.
- Tronstad, R., L. S. Huthoefer, and E. Monke. 1992. "Market Windows and Hedonic Price Analyses: An Application to the Apple Industry." *Journal of Agricultural and Resource Economics* 17: 314–22.
- Tyrvainen, L., and A. Miettinen. 2000. "Property Prices and Urban Forest Amenities." *Journal of Environmental Economics and Management* 39: 205–23.
- Unnevehr, L. J., and S. Bard. 1993. "Beef Quality: Will Consumers Pay for Less Fat?" *Journal of Agricultural and Resource Economics* 18/2: 288–95.
- Viscusi, W. K. 1993. "The Value of Risks to Life and Health." *Journal of Economic Literature* 31/4 (Dec.), 1912–46.
- Ward, C. E., J. L. Lusk, and J. M. Dutton. 2008. "Implicit Value of Retail Beef Product Attributes." *Journal of Agricultural and Resource Economics* 33/3: 364–81.
- Watkins, K. 1999. "Property Valuation and the Structure of Urban Housing Markets." *Journal of Property Investment and Finance* 17: 157–75.
- Waugh, F. 1928. "Quality Factors and Vegetable Prices." *Journal of Farm Economics* 10: 185–96.
- White, H. 1980. "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity." *Econometrica* 48/4: 817–38.
- Winfree, J. A., and J. J. McCluskey. 2005. "Collective Reputation and Quality." *American Journal of Agricultural Economics* 87/1 (Feb.),

206–13.

Zarembka, P. 1974. *Transformation of Variables in Econometrics*. Gottingen: Ibero-Amerika Institut für Wirtschaftsforschung, University of Gottingen.

Notes:

- (1) In Nerlove's article, consumers' preferences are represented by an index function mapping characteristics into valuations of quality, $A(z)$, their exogenously determined prices $P(z)$, and all the other available goods, x . The utility function is represented as $U[V[A(z), Q(z)]; x]$. Given the separability assumptions implied by the commas and an overall utility function homothetic in V versus other goods, the relative prices of the other goods will not influence choices between the varieties, allowing one to keep the focus on the trade-offs between attribute contents and their prices. From utility maximization Nerlove derives a spectrum of demand functions, which takes the form $Q(z) = \phi[P(z); y_1 a(z)]$. Here, y_1 represents consumers' characteristics, which include income, and the quality index $A(z)$ (the consumers' preferences) is represented by a common function of the elements of z , $a(z)$.
- (2) If $y = \exp(\mathbf{X}\boldsymbol{\beta} + \varepsilon)$ and ε is distributed $N(0, \sigma^2)$, then y is log-normally distributed with mean $\exp(\mathbf{X}\boldsymbol{\beta} + 0.5\sigma^2)$ and variance $\exp(2\mathbf{X}\boldsymbol{\beta})[e^{2\sigma^2} - e^{\sigma^2}]$. Taking the log transformation yields the linearized regression model $\ln(y) = \mathbf{X}\boldsymbol{\beta} + \varepsilon$; see Mittelhammer, Judge, and Miller (2000: 209).
- (3) Note that the use of Box-Cox transformation is one of the rare instances in which a fully parametric hedonic model is specified and estimated via maximum likelihood.
- (4) Note that if y is log-normally distributed, estimating $y = \mathbf{X}\boldsymbol{\beta} + \varepsilon$ will produce this kind of heteroskedasticity.
- (5) Since the RESET test is insensitive to heteroskedasticity, a significant RESET test implies misspecification, while heteroskedasticity is signaled by an insignificant RESET and a significant GQ test.
- (6) It should be noticed that the power of a test is increasing in sample size. In the limit, if the sample is large enough, a formal test will reject virtually any hypothesis stated in the form of strict equality.
- (7) There is also the mitigating issue of the need to discover the correct heteroskedastic structure of the error process.
- (8) Koenker 2005: 47), discussing this practice, writes, "You can do it, but it would be wrong," citing a Nixon dictum.
- (9) If $h(p)$ is strictly convex, then $E[h(p)] > h[E(p)]$; if $h(p)$ is strictly concave, then $E[h(p)] < h[E(p)]$. Thus, exponentiation after the logarithmic transformation will tend to underestimate the conditional mean.
- (10) Kernel regression and the Nadaraya-Watson estimator are presented in detail in most graduate-level econometric books. See, e.g., Mittelhammer, Judge, and Miller (2000: 613).
- (11) Common choices are the Epanechnikov and the Tricube Kernel functions.
- (12) Note that Nadaraya-Watson-type estimators can be conceptualized as zero-order polynomial expansions, in which only the intercept $\hat{a}(z_0)$ is estimated.
- (13) (Racine and Li 2004) present a method to include ordered and unordered qualitative regressors in the non-parametric part of the model. See Parmeter, Henderson, and Kumbhakar (2007) for an application to hedonic modeling.

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